

Software Estimation

Software Project Estimation is the **process of estimating various resources required for the completion of a project.**

Estimation is the process of finding an estimate, or approximation, which is a value that can be used for some purpose even if input data may be incomplete, uncertain, or unstable.

Effective software project estimation is one of the most challenging and important activities in software development. Proper project planning and control is not possible without a reliable estimate. The software industry doesn't estimate projects well and doesn't use estimates appropriately.

The four basic steps in software project estimation are:

- 1) **Estimate the size of the development product.** The size can be estimated by using either Lines of Code (LOC) or Function Points (FP).
- 2) **Estimate the effort** in person-months or person-hours (man-month or man-hour). Man-month is an estimate of personal resources required for the project.
- 3) **Estimate the schedule** in calendar days /months/ years based on total man-month required and manpower allocated to the project.
- 4) **Estimate the project cost** in local currency.

COCOMO (Constructive Cost Model)

The Constructive Cost Model or COCOMO Model estimates the cost for any software development project based on the empirical data obtained from observations and uses a mathematical formula to estimate these parameters.

Software projects under COCOMO model strategies are classified into 3 categories, organic, semi-detached, and embedded.

Organic: A software project is said to be an organic type if-

- Project is small and simple.
- Project team is small with prior experience.
- The problem is well understood and has been solved in the past.
- Requirements of projects are not rigid, such a mode example is payroll processing system.

Semi-Detached Mode: A software project is said to be a Semi-Detached type if-

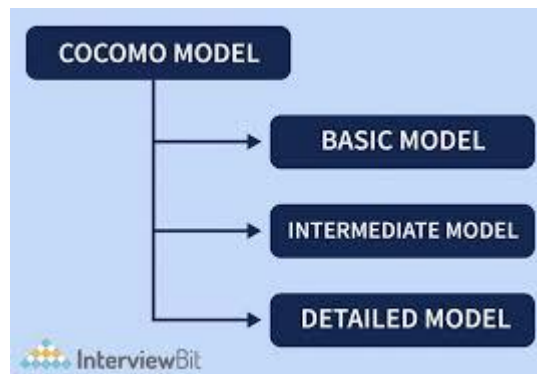
- Project has complexity.
- Project Team will contain both experienced and inexperienced members.
- The project has an intermediate size and has mixed rigid requirements such a mode example is a transaction processing system which has fixed requirements.
- Few such projects are- Database Management System(DBMS), new unknown operating system, difficult inventory management system.

Embedded Mode: A software project is said to be an Embedded mode type if-

- A software project has fixed requirements of resources .
- Product is developed within very tight constraints.
- A software project requiring the highest level of complexity, creativity, and experience requirement fall under this category.

- Such mode software requires a larger team size than the other two models.

Three types of COCOMO models:



Basic COCOMO Model

This COCOMO Model is the simplest form that provides a rough estimate of any project based on the size of the software alone. Hence, it is suitable for small to medium-sized projects with straightforward requirements. Basic COCOMO can be used for quick and slightly rough calculations of Software Costs. This is because the model solely considers based on lines of source code together with constant values obtained from software project types rather than other factors which have major influences on the Software development process as a whole. It requires to calculate the efforts which are required to develop in three modes of development that are organic mode, semi-detached mode, and embedded mode.

The basic COCOMO estimation model is given by the following expressions:

$$E = a (KLOC)^b \text{ person-months}$$

$$D = c (E)^d \text{ months}$$

$$P = E/D \text{ persons}$$

$$\text{Productivity} = KLOC/E \text{ LOC per Person-month}$$

Where,

E is effort applied in person-months.

D is development time in months.

P is the total no. of persons required to accomplish the project.

Productivity= How many lines of code a person can write in one month.

The constant values a,b,c, and d for the Basic Model for the different categories of the system.

Software Project	a	b	c	d
Organic	2.4	1.05	2.5	0.38
Semi-detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

Practice Problems

1. A software project requires 1000 lines of code for development. The project is categorized as **organic type**. Using the basic COCOMO model, calculate the estimated effort, development time, number of persons required to complete the project. Also calculate productivity of it.
2. A software system has 1500 lines of code. The project is classified as **semi-detached**. Using the basic COCOMO model, calculate the estimated effort, development time, number of persons required to complete the project. Also calculate productivity of it.
3. A company is estimating the effort for a new software system with 2000 lines of code. The system is of **embedded** type. Using the basic COCOMO model, calculate the estimated effort, development time, number of persons required to complete the project. Also calculate productivity of it.